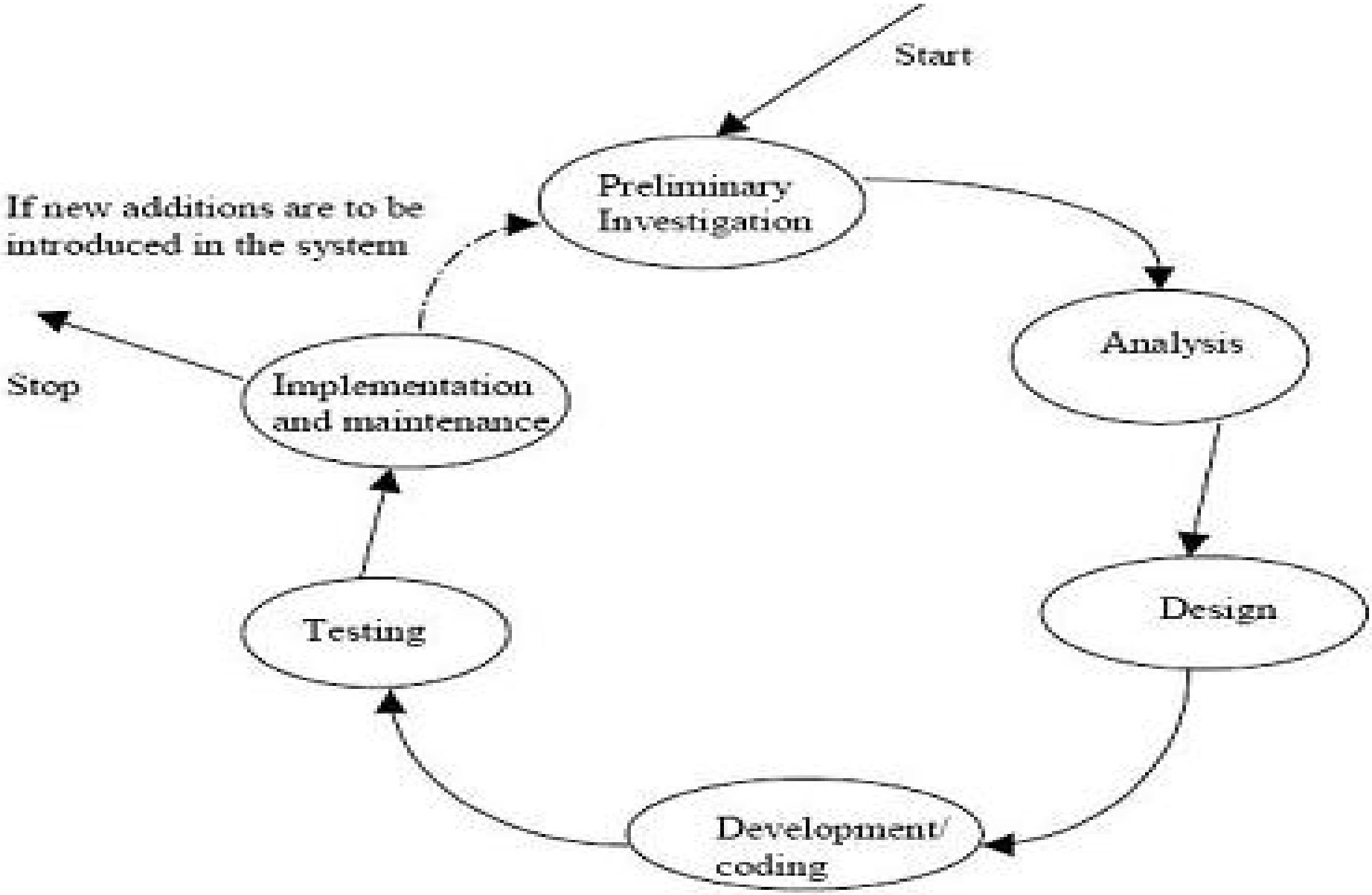
Software development life cycle stages

Stages in building an improved system



Preliminary investigation

- First stage is the preliminary analysis. The main aim of preliminary analysis is to identify the problem. First, need for the new or the enhanced system is established. Only after the recognition of need, for the proposed system is done then further analysis is possible.
- Suppose in an office all leave-applications are processed manually. Now this company is recruiting many new people every year. So the number of employee in the company has increased. So manual processing of leave application is becoming very difficult. So the management is considering the option of automating the leave processing system. If this is the case, then the system analyst would need to investigate the existing system, find the limitations present, and finally evaluate whether automating the system would help the organization.
- Once the initial investigation is done and the need for new or improved system is established, all possible alternate solutions are chalked out. All these systems are known as "candidate systems". All the candidate systems are then weighed and the best alternative of all these is selected as the solution system, which is termed as the "proposed system". The proposed system is evaluated for its feasibility. Feasibility for a system means whether it is practical and beneficial to build that system.
- Feasibility is evaluated from developer and customer's point of view. Developer sees whether they have the required technology or manpower to build the new system. Is building the new system really going to benefit the customer. Does the customer have the required money to build that type of a system? All these issues are covered in the feasibility study of the system. The feasibility of the system is evaluated on the three main issues: technical, economical, and operational. Another issue in this regard is the legal feasibility of the project.

Analysis

Determination of System's requirements: Analysis phase

- After preliminary investigation, analysis phase begins. Analysis is a detailed study of the various operations performed by the system, relationships among the various sub-systems or functional units and finally the relationships outside the system. Major questions under consideration during analysis are:
  1. **What is being done?**
  2. **How is it being done?**
  3. **Does a problem exist?**
  4. **If a problem exists, how severe is it?**
  5. **How frequently does it occur?**
  6. **What is the underlining cause for the problem?**
- Study is conducted to find user's information requirements. Proper functioning of the current system is also studied. Many tools are used during analysis. Data flow diagrams, on-site observations, and questionnaires are some examples.
- Once the analysis is completed, the analyst has a firm knowledge of what is to be done.

Design

- After the system has been analyzed by the analyst, the design stage of system life cycle begins. In design phase, the structure or design for the proposed system is finalized. Structure of files, databases, input, output, processes, and screens(interfaces) are decided. After design is finalized, it is clearly documented in what is called Design Document.
- This design document contains various graphical representations of reports, user interaction screens, etc. This design document is referred by developers during development of system.
- Correct designing of the system is very crucial. If we have a wrong design we won't be able to a get correct desired system. From various studies it is observed that nearly 50% errors are made during design phase in any software development while 33% errors are pertaining to program logic and only 17% are syntactical errors. Also cost of fixing errors is maximum for design phase. Cost of fixing errors for design phase nearly amounts to 80% while it is only 20% for logic and programming.
- Cost of change increases when errors are discovered in later stages for system development. Mostly design errors are discovered in development, testing, or maintenance phase. So the cost of change due to changes in design (which might happen due to some error or improper design is very high. So design process should be handled very carefully.

Development/Coding

- In this phase, the actual development of the system takes place. That is, design representations are translated into actual programs. Software developers may install (or modify and then install) purchased software or they may write new, custom-designed programs.
- Programmers are also responsible for documenting the program, providing an explanation of how procedures are coded. Documentation is essential to test the program and carry on maintenance once the application has been installed. It is also helpful to user in knowing the system well.

System Testing

- After a system has been developed, it is very important to check if it fulfills the customer requirements. For this purpose, testing of the system is done. For testing the systems, various test cases are prepared. A test case is a certain made up situation on which system is exposed so as to find the behavior of system in that type of real situation. These test cases require data. The data can be also made up artificial data or the real data provided by the user.
- There are various types of tests which are used to test the system. These include unit, integration, and acceptance testing.
- The smallest unit of software design is module. In unit testing these modules are tested. Since the modules are very small even individual programmer can test them. Once the individual modules are tested, these are integrated to build the complete system. But testing individual module doesn't guarantee if the system will work properly when these units are integrated.
- Acceptance testing ensures that the system meets all the requirements. If it fulfils the needs then the system is accepted by the customer and put into use.

Implementation and Maintenance

- Implementation of system means putting up system on user's site. Like any system, there is an aging process. Therefore, the system requires periodic maintenance.
- Maintenance can be for software or hardware. User priorities, changes in organizational requirements, or environmental factors call for system enhancements. This is very crucial for the system's life.